Sheaf Theory

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June 20, 2025

Abstract

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1 The General Theory of Sheaves

1.1 Basic Definition of Presheaves

As with how we equipped to each variety V its coordinate ring $k[V]$ which are functions on V, we want to equip to each spectrum some ring which are functions on them. It will not make sense that we can define functions on spectrums immediately.

In its full generality, sheaves are designed to encode all local information into one compact notation. It often has uses in complex geometry for its differential forms and smooth functions. In algebraic geometry it is fundamental for redefining the notion of a variety.

Definition 1.1.1 (The Category of Open Sets) Let $(X,\mathcal{T})$ be a topological space. Define the category $\mathbf{Open}(X)$ to consist of the following data.
- The objects $\text{Ob}(\mathbf{Open}(X))=\mathcal{T}$ are the open sets of X
- For $U,V\subseteq X$ two open sets of X, the morphisms

$$\text{Hom}_{\mathbf{Open}(X)}(U,V)=\begin{cases}\{\iota:U\hookrightarrow V\text{ the inclusion map}\} & \text{if } U\subseteq V\\ \emptyset & \text{otherwise}\end{cases}$$

- Composition is given as the composition of functions.

Recall from Category Theory 1 that a presheaf on $\mathcal{C}$ is just a contravariant functor $\mathcal{C}\to\mathbf{Set}$. Using the category of open sets we can now define presheaves on spaces.

Definition 1.1.2 (Presheaves on a Space) Let X be a space. A presheaf on X is a contravariant functor

$$\mathcal{F}:\mathbf{Open}(X)\to\mathbf{Set}$$

with potentially additional structures on the sets such as $\mathbf{Grp}$, $\mathbf{Ring}$ etc.

Definition 1.1.3 (The Stalk of a Presheaf) Let $\mathcal{F}$ be a presheaf on a topological space $(X,\mathcal{T})$. Let $p\in X$. Consider the full subcategory $\mathcal{J}_p$ of $\mathbf{Open}(X)$ consisting of open sets in X that contain p. Define the stalk of $\mathcal{F}$ at p to be the colimit

$$\mathcal{F}_{X,p}=\underset{\mathcal{J}_p}{\text{colim}}\mathcal{F}(V_p)$$

Lemma 1.1.4
Let X be a topological space. Let $\mathcal{F}$ be a presheaf on X. Let $p\in X$. Then we have

$$\mathcal{F}_{X,p}=\frac{\{(U,f)\mid U\subseteq X\text{ is an open neighborhood of }p, f\in\mathcal{F}(U)\}}{\sim}$$

where $(U,f)\sim(V,g)$ if there exists $W\subseteq U\cap V$ an open neighborhood of p such that $f|_W=g|_W$.

Think of the definition of stalks as follows: Treat f and g to be sections in $\mathcal{F}(U_1)$ and $\mathcal{F}(U_2)$ where $V\subseteq U_1\cap U_2$ is open and contains x. Then we think of f and g to be the same function in the stalk as long as they agree on some open set that contains x.

Indeed, since we do not care about the entirety of the domain of f and g, and only care about what happens locally near x, it makes sense for us to treat them as a function when they appear to be the same locally.

Definition 1.1.5 (Morphism of Presheaves)
Let X be a topological space. Let $\mathcal{F},\mathcal{G}$ be presheaves on X. A morphism $\varphi:\mathcal{F}\to\mathcal{G}$ is a natural transformation of functors.

Notice that the natural transformation ϕ here takes every open set U and maps it to a group homomorphism $\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ if $\mathcal{F}$ and $\mathcal{G}$ are presheaves with values in **Grp**.

Definition 1.1.6 (Induced Map of Stalks) Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be presheaves on X. Let $p \in X$. Define the induced map of stalks

$$\phi_p : \mathcal{F}_{X,p} \to \mathcal{G}_{X,p}$$

to be the map induced by the universal product of colimits.

Definition 1.1.7 (Isomorphism of Presheaves)
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be presheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is an isomorphism if φ is a natural isomorphism. In this case, we say that $\mathcal{F}$ and $\mathcal{G}$ are isomorphic.

1.2 Basic Definition of Sheaves

Definition 1.2.1 (Sheaves) Let X be a space. A sheaf on X is a presheaf $\mathcal{F} : \textbf{Open} \to \textbf{Sets}$ satisfying two additional properties

- Identity: If $\{U_i | i \in I\}$ is an open cover of U and $\phi_1, \phi_2 \in \mathcal{F}(U)$ and $\phi_1|_{U_i} = \phi_2|_{U_i}$ for all i, then $\phi_1 = \phi_2$
- Gluing: If $\{U_i | i \in I\}$ is an open cover of U and $\phi_i \in \mathcal{F}(U_i)$ for all $i \in I$ such that $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is exists some $\phi \in \mathcal{F}(U)$ such that $\phi|_{U_i} = \phi_i$ for all $i \in I$.

Proposition 1.2.2
Let X be a space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. Then φ is an isomorphism if and only if for all $p \in X$, the induced map

Proof Suppose that ϕ is an isomorphism of sheaves. Then we obtain the following commutative diagram:

$$\begin{tikzcd}
 & \mathcal{F}_{X,p} \arrow[d, dashed, "\exists! a"] & \\
\mathcal{F}(U) \arrow[ur] \arrow[d, "\phi(U)"'] & \mathcal{G}_{X,p} & \mathcal{F}(V) \arrow[ul] \arrow[d, "\phi(V)"] \\
\mathcal{G}(U) \arrow[ur] \arrow[rr, "\text{res}_{V,U}"] & & \mathcal{G}(V) \arrow[ul]
\end{tikzcd}$$

which is due to the fact that $\phi(U)$ and $\phi(V)$ are isomorphisms. This unique map $a : \mathcal{G}_{X,P} \to \mathcal{F}_{X,p}$ exists by the universal property of the colimit. Similarly, we can construct a map $b : \mathcal{F}_{X,P} \to G_{X,p}$. Since $\mathcal{F}_{X,p}$ and $\mathcal{G}_{X,p}$ are colimits, they unique map from $\mathcal{F}_{X,p}$ to itself must be the identity. Hence $a \circ b$ is the identity. Similarly, $b \circ a$ is also the identity hence $\mathcal{F}_{X,p}$ and $\mathcal{G}_{X,p}$ are isomorphic.

Now suppose that ϕ_p is an isomorphism for each $p \in X$. We show that $\phi(U)$ is bijective for all $U \subseteq X$ open. Let $s \in \mathcal{F}(U)$ such that $\phi(U)(s) = 0$. Then for each $p \in U$, $\phi(U)(s)$ considered as an element in $\mathcal{G}_{X,p}$ is equal to 0. Since ϕ_p is injective, $s = 0$ in $\mathcal{F}_{X,p}$. This means that there exists an open neighbourhood $V_p \subseteq U$ of p such that $s \in \phi(V_p)$ is 0 for each $p \in U$. Since U is covered by the open neighbourhoods V_p for $p \in U$, by the identity axiom we conclude that $s = 0$ in $\phi(U)$. Thus ϕ is injective.

Now let $t \in \mathcal{G}(U)$. Let t_p be the corresponding element of t in $\mathcal{G}_{X,p}$ for each $p \in U$. Since ϕ_p is surjective, there exists $s_p \in \mathcal{F}_{X,p}$ such that $\phi_p(s_p) = t_p$ for each $p \in U$. Suppose that on a neighbourhood V_p of p, s_p is represented as r_p. Then $\phi(r_p) \in \mathcal{G}(V_p)$ and $t|_{V_p}$ are two elements whose germs in $\mathcal{G}_{X,p}$ are equal. Hence there exists $W_p \subseteq V_p$ containing p such that $\phi(r_p) = t|_{W_p}$ in $\mathcal{G}(W_p)$. Now U is covered by open sets of the form W_p for each $p \in U$. Let $p \in W_p$ and $q \in W_q$. Then both $r_p|_{W_p \cap W_q}$ and $r_q|_{W_p \cap W_q}$ are two sections in $\mathcal{F}(W_p \cap W_q)$ that are sent to $t|_{W_p \cap W_q} \in \mathcal{G}(W_p \cap W_q)$

by $\phi(W_p \cap W_q)$. By injectivity, they are equal. We conclude that

$$r_p|_{W_p \cap W_q} = r_q|_{W_p \cap W_q}$$

By the gluing axiom, we conclude that there exists a section $s \in \mathcal{F}(U)$ such that $s|_{W_p} = r_p$. Now $\phi(s)$ and t are two sections in $\mathcal{G}(U)$ such that for any $p \in U$, $\phi(s)|_{W_p} = t|_{W_p}$. By the identity axiom, we conclude that $\phi(s) = t$. ■

Note that the above proposition is very untrue for presheaves because as one can see, proving both injective and surjective requires to use of the sheaf axioms instead of just the presheaf datum.

Definition 1.2.3 (Sheafification) Let X be a space. Let $\mathcal{F}$ be a presheaf on X. The sheafification of $\mathcal{F}$ consists of a sheaf $\mathcal{F}^+$ and a morphism of presheaf $\theta : \mathcal{F} \to \mathcal{F}^+$ such that the following universal property is satisfied. For any sheaf G on X and morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$, there exists a unique morphism $\varphi^+ : \mathcal{F}^+ \to \mathcal{G}$ such that the following diagram commutes:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\theta"] \arrow[dr, "\varphi"'] & \mathcal{F}^+ \arrow[d, dashed, "\exists!\varphi^+"] \\ & \mathcal{G} \end{tikzcd}$$

Proposition 1.2.4 Let X be a space and $\mathcal{F}$ a presheaf on X. Then $\mathcal{F}^+$ exists and is unique up to isomorphism. Explicitly, $\mathcal{F}^+$ can be defined as follows.

- For each open set $U \subseteq X$,

$$\mathcal{F}^+(U) = \left\{ s : U \to \bigcup_{p \in U} \mathcal{F}_p \;\middle|\; \substack{\forall p \in U,\ s(p) \in \mathcal{F}_{X,p} \text{ and } \exists V_p \subseteq U \text{ and} \\ t \in \mathcal{F}(V_p) \text{ s.t. } \forall q \in V_p, s(q) = t_q \in \mathcal{F}_{X,q}} \right\}$$

- For each $V \subseteq U$ an inclusion, there is a unique morphism $\mathcal{F}^+(U) \to \mathcal{F}^+(V)$ that sends $s \in U$ to its restriction

$$s|_V : V \to \bigcup_{p \in V} \mathcal{F}_{X,p}$$

We end the section with a formula to construct a sheaf given its values on just the basis of a space.

Proposition 1.2.5 Let X be a topological space. Let $\mathcal{B}$ be the basis of X. Suppose that $\mathcal{F}_0$ is a sheaf defined on the basis $\mathcal{B}$ of X. Then the natural extension to open sets U by

$$\mathcal{F}(U) = \left\{ (s_i)_i \in \prod_i \mathcal{F}_0(B_i) \;\middle|\; B_i \in \mathcal{B}, B_i \subseteq U, s_i|_{B_i \cap B_j} = s_j|_{B_i \cap B_j} \right\} = \varprojlim_{\substack{B \in \mathcal{B} \\ B \subseteq U}} \mathcal{F}(B)$$

defines a sheaf for X.

Proof ■

This means that sheaves are uniquely determined by their values in the basis of X. We can simply define the sheaf on the basis elements and by this natural extension, a sheaf will be defined for all of X.

1.3 Sheaves with Values in a Category

Presheaves are easily generalized to take values in an arbitrary category because they are simply defined to be contravariant functors.

Definition 1.3.1 (Presheaves with Values in a Category) Let X be a space. Let $\mathcal{C}$ be a category. A presheaf on X with values in $\mathcal{C}$ is a contravariant functor

$$\mathcal{F} : \textbf{Open}(X) \to \mathcal{C}$$

However, it is hard to define sheaves with values on an arbitrary category $\mathcal{C}$. This is because in our definition of sheaves on sets, we made use of the fact that morphisms in **Set** have a well defined notion of restriction. We will define sheaves on a general category by first considering sheaves with an underlying set.

Theorem 1.3.2 Let X be a space. Let $\mathcal{F}$ be a presheaf with values in **Set**. Then the following are equivalent characterizations.

- $\mathcal{F}$ is a sheaf
- For any open covering $U = \bigcup_{i\in I} U_i$, the sequence
$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \xrightarrow{g} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$
where f is given by the product of the unique maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$ and g is given by sending $(s_i)_{i\in I}$ to $(s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j\in I}$ is exact.
- For any open covering $U = \bigcup_{i\in I} U_i$, the following diagram
$$\mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \underset{h}{\overset{g}{\rightrightarrows}} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$
is an equalizer diagram, where f is given by the product of the unique maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$, g is given by sending $(s_i)_{i\in I}$ to $(s_i|_{U_i\cap U_j})_{i,j\in I}$, h is given by sending $(s_i)_{i\in I}$ to $(s_j|_{U_i\cap U_j})_{i,j\in I}$.

Proof

We first note that the second and third condition are equivalent. Therefore we just have to show that the first condition is equivalent to the third one. Let $\mathcal{F}$ be an abelian sheaf. We want to show that $\text{im}(f) = \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$. Therefore we begin with $x \in \text{im}(f)$. Then $f(x) = (x_i)_{i\in I}$. But we have that $g((x_i)_{i\in I}) = (x_i|_{U_i\cap U_j})_{i,j\in I}$ and $h((x_i)_{i\in I}) = (x_j|_{U_i\cap U_j})_{i,j\in I}$. We want to show that for all $i, j \in I$, we have that $x_i|_{U_i\cap U_j} = x_j|_{U_i\cap U_j}$. But x_i is just the restriction of x to U_i and x_j is the restriction of x to U_j. But by definition of a sheaf, restricting to V_i and further to V_{ij} is the same as directly restricting to V_{ij}. Similarly, restricting to V_j and further to V_{ij} is the same as directly restricting to V_{ij}. We conclude that applying g and h to $(x_i)_{i\in I}$ give the same element. Thus $x \in \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$.

Now let $(y_i)_{i\in I} \in \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$. This means that $y_i|_{U_i\cap U_j} = y_j|_{U_i\cap U_j}$ for all $i, j \in I$. By the gluing axiom, there exists $y \in C(U)$ such that $y|_{U_i} = y_i$. Thus y is in the image of f and we conclude that $y \in \text{im}(f)$.

Finally, we also need to show that f is injective. Suppose that $x, y \in \mathcal{F}(U)$ is such that $x_i = y_i$ for all $i \in I$. Then by the identity axiom we conclude that $x = y$. Thus f is injective. Thus the equalizer condition is satisfied.

Now suppose that the equalizer condition is satisfied. Then f is injective. This means that for $x, y \in \mathcal{F}(U)$ such that $x_i = y_i$ for all $i \in I$, then $x = y$. But this is precisely the identity axiom. Suppose that $\text{im}(f) = \text{Eq}(g, h)$. Let $s_i \in \mathcal{F}(U_i)$ such that

$$s_i|_{U_i\cap U_j} = g(s_i) = h(s_i) = s_j|_{U_i\cap U_j}$$

for all $i, j \in I$. Then $(s_i)_{i\in I} \in \text{Eq}(g, h) = \text{im}(f)$. Which means that there exists some $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. But this is the content of the gluing axiom. Since the identity and the gluing axiom are satisfied, we conclude that $\mathcal{F}$ is a sheaf. ∎

Motivated by the above characterization, we define sheaves with values on a category with finite products as follows.

Definition 1.3.3 (Sheaves with Values in a Category) Let X be a space. Let $\mathcal{C}$ be a category with finite products. Let $\mathcal{F}$ be a presheaf on X with values in $\mathcal{C}$. We say that $\mathcal{F}$ is a sheaf with values in $\mathcal{C}$ if for every open covering $U = \bigcup_{i\in I} U_i$, the following diagram

$$\mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \underset{h}{\overset{g}{\rightrightarrows}} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$

is an equalizer diagram, where f is given by the product of the unique maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$, g is given by sending $(s_i)_{i\in I}$ to $(s_i|_{U_i\cap U_j})_{i,j\in I}$, h is given by sending $(s_i)_{i\in I}$ to $(s_j|_{U_i\cap U_j})_{i,j\in I}$.

2 The Abelian Category of Sheaves

2.1 The Category of Sheaves

Definition 2.1.1 (The Category of Sheaves of Abelian Groups) Let X be a topological space. Define the category $\mathbf{Ab}(X)$ of sheaves on X with values in $\mathbf{Ab}$ is the category defined as follows.

- The objects are the sheaves on X with values in $\mathbf{Ab}$.
- Given two sheaves $\mathcal{F}$ and $\mathcal{G}$ on X, a morphism from $\mathcal{F}$ to $\mathcal{G}$ is a morphism of sheaves $\phi : \mathcal{F} \to \mathcal{G}$ which is just a natural transformation
- Composition is given by the composition of natural transformations

Proposition 2.1.2
Let X be a space. Let $i : \mathbf{Ab}(X) \to \mathbf{PAb}(X)$ be the forgetful functor. Then there is an adjunction

$$i : \mathbf{Ab}(X) \rightleftarrows \mathbf{PAb}(X) : (-)^+$$

Explicitly, there is a natural isomorphism

$$\mathrm{Hom}_{\mathbf{PAb}(X)}(i(\mathcal{F}), \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{Ab}(X)}(\mathcal{F}, \mathcal{G}^+)$$

for $\mathcal{F}$ a sheaf on X and $\mathcal{G}$ a presheaf on X.

Proposition 2.1.3
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are true.

- The (categorical) kernel of φ exists and is given by $\ker(\phi)(U) = \ker(\phi(U))$.
- The (categorical) image of φ exists and is given by $\mathrm{im}(\phi)(U) = \mathrm{im}(\phi(U))$.
- The (categorical) cokernel of φ exists and is given by the sheafification of the presheaf $\mathrm{coker}(\phi)(U) = \mathrm{coker}(\phi)(U)$.

Proposition 2.1.4
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are true.

- $\ker(\phi)_p = \ker(\phi_p)$.
- $\mathrm{im}(\phi)_p = \mathrm{im}(\phi_p)$.

Proposition 2.1.5
Let X be a topological space. Then $\mathbf{Ab}(X)$ is an abelian category.

Definition 2.1.6 (Injective and Surjective Morphisms)
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism.

- We say that φ is injective if $\ker(\varphi) = 0$.
- We say that φ is surjective if $\mathrm{im}(\varphi) = \mathcal{G}$.

Lemma 2.1.7
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then φ is an isomorphism if and only if φ is injective and surjective.

Proposition 2.1.8
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are equivalent.

- φ is injective.
- For each $U \subseteq X$ open, $\varphi(U)$ is injective.
- For each $p \in X$, φ_p is injective.

Proposition 2.1.9
Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are equivalent.
- φ is surjective.
- For each $U \subseteq X$ open and for every $g \in \mathcal{G}(U)$, there exists an open cover $\{U_i \mid i \in I\}$ of U and there exists $f_i \in \mathcal{F}(U_i)$ such that $\varphi(U_i)(f_i) = g|_{U_i}$.
- For each $p \in X$, φ_p is surjective.

2.2 Functoriality of the Category of Sheaves

Definition 2.2.1 (Direct Image Sheaf)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{F}$ be a sheaf on X. Define the direct image of sheaf $f_*\mathcal{F} : \mathbf{Open}(Y) \to \mathbf{Set}$ as follows.
- For each open set $U \subseteq Y$, define $f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$.
- For each inclusion $V \subseteq U$, define the induced map
$$\mathcal{F}(f^{-1}(U)) \to \mathcal{F}(f^{-1}(V))$$
to be the map $\mathcal{F}(\iota : f^{-1}(U) \to f^{-1}(V))$.

Proposition 2.2.2
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{F}$ be a sheaf on X. Then $f_*\mathcal{F}$ is a sheaf on Y.

Definition 2.2.3 (The Direct Image Sheaf Functor)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{F}$ be a sheaf on X. Define the direct image sheaf functor
$$f_* : \mathbf{Ab}(X) \to \mathbf{Ab}(Y)$$
by the following data.
- For each sheaf $\mathcal{F}$ of X, $f_*\mathcal{F}$ is the direct image sheaf on Y.
- For each morphism $\varphi : \mathcal{F} \to \mathcal{G}$, $f_*\varphi : f_*\mathcal{F} \to f_*\mathcal{G}$ is the morphism given by
$$\varphi(f^{-1}(U)) : \mathcal{F}(f^{-1}(U)) \to \mathcal{G}(f^{-1}(U))$$

Definition 2.2.4 (Inverse Image Sheaf)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{G}$ be a sheaf on Y. Define the inverse image sheaf $f^{-1}\mathcal{G}$ to be the sheafification of the presheaf on X defined as follows.
- For each open set $U \subseteq X$, let $\mathcal{J}(U) \subseteq \mathbf{Open}(X)$ be the full subcategory consisting of open sets of Y the contain $f(U)$. Then the presheaf sends U to $\lim_{W \in \mathcal{J}(U)} \mathcal{G}(W)$.
- For each inclusion $V \subseteq U$, define the induced map to be induced by the universal property of the limit.

Definition 2.2.5 (The Inverse Image Sheaf Functor)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{F}$ be a sheaf on X. Define the inverse image sheaf functor
$$f^{-1} : \mathbf{Ab}(Y) \to \mathbf{Ab}(X)$$
by the following data.
- For each sheaf $\mathcal{F}$ of Y, $f^{-1}\mathcal{F}$ is the inverse image sheaf on X.
- For each morphism $\varphi : \mathcal{F} \to \mathcal{G}$, the morphism $f^{-1}\mathcal{F} \to f^{-1}\mathcal{G}$ is induced by the universal property of limits.

Proposition 2.2.6
Let X, Y be spaces. Let $f : X \to Y$ be a map. Then there is an adjunction
$$f^{-1} : \mathbf{Ab}(Y) \rightleftarrows \mathbf{Ab}(X) : f_*$$

Explicitly, there is a natural isomorphism

$$\mathrm{Hom}_{\mathbf{Ab}(X)}(f^{-1}\mathcal{G},\mathcal{F})\cong\mathrm{Hom}_{\mathbf{Ab}(Y)}(\mathcal{G},f_*\mathcal{F})$$

2.3 Ringed Spaces

Definition 2.3.1 (Ringed Space)
A ringed space is a topological space X with a sheaf of rings $\mathcal{O}_X$ on X.

Definition 2.3.2 (Morphisms of Ringed Spaces) Let $(X,\mathcal{O}_X)$ and $(Y,\mathcal{O}_Y)$ be ringed spaces. A morphism of ringed spaces from $(X,\mathcal{O}_X)$ to $(Y,\mathcal{O}_Y)$ is a pair $(f,f^\#)$ of continuous map $f:X\to Y$ and a map $f^\#:\mathcal{O}_Y\to f_*\mathcal{O}_X$ of sheaves of rings on Y.

Suppose that $(X,\mathcal{O}_X)$ and $(Y,\mathcal{O}_Y)$ are locally ringed spaces. Given a point $p\in X$, we obtain induced maps

$$f^\#:\mathcal{O}_Y(V)\to f_*\mathcal{O}_X(f^{-1}(V))$$

for any neighbourhood V of $f(p)$. Taking the colimit, we obtain a map

$$\mathcal{O}_{Y,f(p)}=\mathrm{colim}_V\mathcal{O}_Y(V)\to\mathrm{colim}_V\mathcal{O}_X(f^{-1}(V))$$

As V ranges through the neighbourhood of $f(p)$, $f^{-1}(V)$ ranges over a subset of the neighbourhoods of p so that we obtain a map

$$\mathcal{O}_{Y,f(p)}=\mathrm{colim}_V\mathcal{O}_Y(V)\to\mathrm{colim}_V\mathcal{O}_X(f^{-1}(V))\to\mathcal{O}_{X,p}$$

Definition 2.3.3 (Locally Ringed Spaces)
Let $(X,\mathcal{O}_X)$ be a ringed space. We say that X is a locally ringed space if for all $p\in X$, $\mathcal{O}_{X,p}$ is a local ring.

Definition 2.3.4 (Morphisms of Locally Ringed Spaces) Let $(X,\mathcal{O}_X)$ and $(Y,\mathcal{O}_Y)$ be locally ringed spaces. Then a morphism of locally ringed spaces is a morphism of ringed spaces such that for each $p\in X$, the induced map of local rings

$$f_p^\#:\mathcal{O}_{Y,f(p)}\to\mathcal{O}_{X,p}$$

is a local homomorphism of local rings.

Definition 2.3.5 (The Category of Locally Ringed Spaces)
Define the category **LRS** of locally ringed spaces as follows.
- The objects are locally ringed spaces.
- The morphisms are morphisms of locally ringed spaces.
- Composition is given by the composition of continuous functions and composition of natural transformations.

3 Modules over a Sheaf

3.1 The Category of $\mathcal{O}_X$-Modules

Definition 3.1.1 (Sheaf of $\mathcal{A}$-modules) Let $(X, \mathcal{A})$ be a ringed space. We say that an abelian sheaf $\mathcal{F}$ of X is a sheaf of $\mathcal{A}$-modules if the following are true.

- For every open set $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{A}(U)$-module
- For each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathcal{F}(U) \to \mathcal{F}(V)$ is such that the following diagram commutes:

$$\begin{tikzcd} \mathcal{A}(U) \times \mathcal{F}(U) \arrow[r, "\text{action}"] \arrow[d, "\text{res}_{U,V} \times \text{res}_{U,V}"'] & \mathcal{F}(U) \arrow[d, "\text{res}_{U,V}"] \\ \mathcal{A}(V) \times \mathcal{F}(V) \arrow[r, "\text{action}"] & \mathcal{F}(V) \end{tikzcd}$$

Definition 3.1.2 (The Category of $\mathcal{A}$-Modules) Let $(X, \mathcal{O})$ be a ringed space. Define the category of $\mathcal{O}_X$-modules $\mathbf{Mod}_{\mathcal{O}_X}$ to consist of the following data.

- The objects consists sheaves of $\mathcal{O}_X$-modules.
- The morphisms are just morphisms of sheaves between $\mathcal{O}_X$-modules.
- Composition is given by the composition of morphisms of sheaves.

Proposition 3.1.3
Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathbf{Mod}_{\mathcal{O}_X}$ is an abelian category.

3.2 Functoriality of the Category of Modules over a Sheaf

Definition 3.2.1 (The Direct Image Module)
Let $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$ be ringed spaces. Let $f : X \to Y$ be a morphism. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Define the direct image of $\mathcal{F}$ to be the $\mathcal{O}_Y$-module $f_*\mathcal{F}$ whose module structure is given by the morphism $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ and the $f_*\mathcal{O}_X$-module structure of $f_*\mathcal{F}$.

Definition 3.2.2 (The Direct Image Module Functor)

Definition 3.2.3 (The Inverse Image Module)
Let $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$ be ringed spaces. Let $f : X \to Y$ be a morphism. Let $\mathcal{G}$ be an $\mathcal{O}_Y$-module. Define the inverse image module of $\mathcal{G}$ to be

$$f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

Definition 3.2.4 (The Inverse Image Module)

Proposition 3.2.5 There is an adjunction given by

$$f^* : \mathbf{Mod}_{\mathcal{O}_Y} \rightleftarrows \mathbf{Mod}_{\mathcal{O}_X} : f_*$$

Explicitly, there is a natural isomorphism

$$\text{Hom}_{\mathbf{Mod}_{\mathcal{O}_X}}(f^*\mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\mathbf{Mod}_{\mathcal{O}_Y}}(\mathcal{G}, f_*\mathcal{F})$$

for $\mathcal{F}$ an $\mathcal{O}_X$-module and $\mathcal{G}$ an $\mathcal{O}_Y$-module.

3.3 Quasi-Coherent Sheaves

Quasi-coherent sheaves and coherent sheaves has extremely abstract definitions. If one is only interested in the case of Algebraic Geometry, then there is a very nice explicit way of defining it, so readers may skip this part.

Definition 3.3.1 (Quasi-Coherent Sheaves) Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-module. We say that $\mathcal{F}$ is a quasi-coherent sheaf if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an exact sequence:

$$\mathcal{O}_X^{\otimes I}|_U \longrightarrow \mathcal{O}_X^{\otimes J}|_U \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

for some countable indexing sets I and J.

Definition 3.3.2 (The Category of Quasi-Coherent Sheaves) Let $(X, \mathcal{O}_X)$ be a ringed space. Define the category of quasi-coherent sheaves

$$\mathbf{QCoh}_{\mathcal{O}_X}$$

over $(X, \mathcal{O}_X)$ as follows.

- The objects are the quasi-coherent sheaves $\mathcal{F}$ over $(X, \mathcal{O}_X)$
- The morphisms are morphisms between sheaves of modules. This means that if $\mathcal{F}$ and $\mathcal{G}$ are quasi-coherent $\mathcal{O}_X$-modules, then a morphism $\mathcal{F} \to \mathcal{G}$ is a natural transformation $\lambda : \mathcal{F} \to \mathcal{G}$ that respects the module structure.
- Composition is given by the composition of sheaves

Proposition 3.3.3
Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathbf{QCoh}_{\mathcal{O}_X}$ is an abelian category.

3.4 Coherent Sheaves

Definition 3.4.1 (Sheaf of Finite Type) Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-module. We say that $\mathcal{F}$ is a sheaf of finite type if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is a surjective morphism

$$\mathcal{O}_X^{\otimes n}|_U \to \mathcal{F}|_U$$

for some $n \in \mathbb{N}$.

Definition 3.4.2 (Coherent Sheaves) Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-module. We say that $\mathcal{F}$ is coherent if the following are true.

- $\mathcal{F}$ is a sheaf of finite type.
- For any $U \subseteq X$ and any morphism

  $$\varphi : \mathcal{O}_X^{\otimes n}|_U \to \mathcal{F}|_U$$

  of $\mathcal{O}_X$-modules, then kernel of φ is a sheaf of finite type.

Definition 3.4.3 (The Category of Quasi-Coherent Sheaves) Let $(X, \mathcal{O}_X)$ be a ringed space. Define the category of quasi-coherent sheaves

$$\mathbf{Coh}_{\mathcal{O}_X}$$

over $(X, \mathcal{O}_X)$ as follows.

- The objects are the coherent sheaves $\mathcal{F}$ over $(X, \mathcal{O}_X)$
- The morphisms are morphisms between sheaves of modules. This means that if $\mathcal{F}$ and $\mathcal{G}$ are quasi-coherent $\mathcal{O}_X$-modules, then a morphism $\mathcal{F} \to \mathcal{G}$ is a natural transformation $\lambda : \mathcal{F} \to \mathcal{G}$ that respects the module structure.
- Composition is given by the composition of sheaves

Theorem 3.4.4 Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathbf{Coh}_{\mathcal{O}_X}$ is an abelian category.

Proposition 3.4.5 Let $(X, \mathcal{O}_X)$ be a ringed space. Let the following
$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}'' \longrightarrow 0$$
be an exact sequence of sheaves. If two of the three sheaves are coherent, then so is the last one.

3.5 Locally Free Sheaves

Definition 3.5.1 (Free and Locally Free Sheaves) Let $(X, \mathcal{A})$ be a ringed space. Let $\mathcal{F}$ be a $\mathcal{A}$-module. We say that $\mathcal{F}$ is free is there is an isomorphism
$$\mathcal{F} \cong \mathcal{A}^{\oplus n}$$
of sheaves. We say that $\mathcal{F}$ is locally free if X has an open cover $\{U_i \mid i \in I\}$ such that for each U_i, there is an isomorphism
$$\mathcal{F}|_{U_i} \cong (\mathcal{A}|_{U_i})^{\oplus n}$$
of sheaves. In this case we say that the rank of $\mathcal{F}$ is n.

Lemma 3.5.2 If X is connected then the rank of a locally free sheaf on X is constant.

Proposition 3.5.3 Let $(X, \mathcal{O})$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Then $\mathcal{F}$ is locally free of rank r if and only if
$$\mathcal{F}_{X,p} \cong \mathcal{O}_{X,p}^r$$
for all $p \in X$.

Definition 3.5.4 (The Category of Locally Free Sheaves) Let $(X, \mathcal{A})$ be a ringed space. Define the category of locally free sheaves $\mathbf{Loc}_n$ of rank n over X as follows.

- The objects are the locally free sheaves of rank n over X
- The morphisms are the morphisms of sheaves of modules
- Composition is given by the composition of morphisms of sheaves.

3.6 Invertible Sheaves

Definition 3.6.1 (Invertible Sheaf) Let X be a ringed space. We say that a sheaf $\mathcal{F}$ on X is invertible if it is a locally free sheaf on rank 1.

Theorem 3.6.2 Let $(X, \mathcal{O}_X)$ be a ringed space. Then the following are equivalent characterization of a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$

- $\mathcal{F}$ is invertible
- There exists a sheaf G such that $F \otimes_{\mathcal{O}_X} G \cong \mathcal{O}_X$
- $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee \cong \mathcal{O}_X$

Proposition 3.6.3 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ and $\mathcal{G}$ be invertible sheaves on X. Then
$$\mathcal{F} \otimes \mathcal{G}$$
is also an invertible sheaf.

Definition 3.6.4 (The Picard Group) Let $(X, \mathcal{O}_X)$ be a ringed space. Define the Picard group of X to be the set
$$\text{Pic}(X) = \{\mathcal{L} \mid \mathcal{L} \text{ is an invertible sheaf on } X\}$$
together with the tensor product $\otimes$.

4 Sheaf Cohomology

4.1 The Global Section Functor

Definition 4.1.1 (Global Section Functor) Let X be a space. Define the global section functor to be the functor $\Gamma(X,-):\mathbf{Ab}(X)\to\mathbf{Ab}$ defined as follows.

- A sheaf $\mathcal{F}$ of groups on X is sent to $\Gamma(X,\mathcal{F})=\mathcal{F}(X)$
- For a morphism of sheaves $\phi:\mathcal{F}\to\mathcal{G}$, $\Gamma(X,-)$ sends it to the ring homomorphism $\phi(X):\mathcal{F}(X)\to\mathcal{G}(X)$

In general, we can define functor by sending every sheaf to a the local section of a chosen open set. This operation in general is left exact. This means that we have the following proposition.

Proposition 4.1.2 Let X be a space and let $\mathcal{F},\mathcal{F}',\mathcal{F}''$ be sheaves on X such that there is an exact sequence

$$0\longrightarrow\mathcal{F}\xrightarrow{\phi}\mathcal{F}'\xrightarrow{\psi}\mathcal{F}''\longrightarrow 0$$

Then there is an exact sequence

$$0\longrightarrow\mathcal{F}(U)\xrightarrow{\phi(U)}\mathcal{F}'(U)\xrightarrow{\psi(U)}\mathcal{F}''(U)$$

of local sections for any chosen open set $U\subseteq X$.

Proof We want to show that $\ker(\phi(U))=0$. Suppose that $s\in\ker(\phi(U))$. Then $\phi(U)(s)=0$. Then for any $p\in U$, $(\phi(U),\phi(U)(s))\in\mathcal{G}_{X,p}$ is equivalent to the 0 section. By 1.4.5, ϕ_p is injective hence s is equivalent to the zero section in $\mathcal{F}_{X,p}$ for some open subset U_p of U. Since this is true for any p, we conclude that $s=0$ in $\mathcal{F}(U)$ by the gluing axiom.

It remains to show that $\operatorname{im}(\phi(U))=\ker(\psi(U))$. Suppose that $t\in\operatorname{im}(\phi(U))$. Then there exists a section $s\in\mathcal{F}(U)$ such that $\phi(U)(s)=t$. By 1.4.6, $\psi(U)(t)=(\psi\circ\phi)(U)(s)=0$ $\mathcal{F}''_{X,p}$ for every $p\in U$. Since this is true for all p, we conclude that $\psi(U)(t)=0$ by gluing axiom. Hence $\operatorname{im}(\phi(U))\subseteq\ker(\psi(U))$.

Now suppose that $t\in\ker(\psi(U))$. Then $\psi(U)(t)=0$ and hence $(\psi(U),\psi(U)(t))$ is equivalent to to the zero section in $\mathcal{F}''_{X,p}$ for every $p\in U$. ■

Corollary 4.1.3 Let X be a space. The global section functor $\Gamma(X,-):\mathbf{Ab}(X)\to\mathbf{Ab}$ is a left exact functor.

Proof The global section functor is a special case of the above proposition. ■

Definition 4.1.4 (Cohomology Functors) Let X be a space. Define the cohomology functors

$$H^i(X,-)=R^i\Gamma(X,-):\mathbf{Ab}(X)\to\mathbf{Ab}$$

of X to be the right derived functors of $\Gamma(X,-):\mathbf{Ab}(X)\to\mathbf{Ab}$. For any sheaf $\mathcal{F}$ of abelian groups on X, the groups $H^i(X,\mathcal{F})$ are called the cohomology groups of $\mathcal{F}$.

4.2 Flasque Sheaves

Definition 4.2.1 (Flasque Sheaves) A sheaf $\mathcal{F}$ on a space X is said to be flasque if for every pair of open sets $V\subset U$, the restriction map $\mathcal{F}(U)\to\mathcal{F}(V)$ is surjective.

Lemma 4.2.2 If $(X,\mathcal{O}_X)$ is a ringed space, then any injective $\mathcal{O}_X$-module is flasque.

Lemma 4.2.3 Let X be a space and let $\mathcal{F}, \mathcal{F}', \mathcal{F}''$ be sheaves on X such that there is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}'' \longrightarrow 0$$

of sheaves. Then the following are true.

- If $\mathcal{F}$ and $\mathcal{F}'$ are flasque, then $\mathcal{F}''$ is flasque.
- If $\mathcal{F}$ is flasque, then for any $U \subseteq X$ open, there is an exact sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\phi} \mathcal{F}'(U) \xrightarrow{\psi} \mathcal{F}''(U) \longrightarrow 0$$

Proof Suppose that $\mathcal{F}$ and $\mathcal{F}'$ are flasque. Then by naturality we obtain a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(U) \arrow[r, "\phi(U)"] \arrow[d] & \mathcal{F}'(U) \arrow[r, "\psi(U)"] \arrow[d] & \mathcal{F}''(U) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F}(V) \arrow[r, "\phi(V)"] & \mathcal{F}'(V) \arrow[r, "\psi(V)"] & \mathcal{F}''(V) \arrow[r] & 0
\end{tikzcd}$$

for any $V \subseteq U$ open sets. Let $s \in \mathcal{F}''(V)$. Then by surjectivity of $\psi(V)$ and $\mathcal{F}'(U) \to \mathcal{F}'(V)$, there exists $t \in \mathcal{F}'(U)$ that maps correspondingly by the same maps to s. Then using the map $\phi(U)$ we can send s to an element $r \in \mathcal{F}''(U)$ such that r is sent to s under the map $\mathcal{F}''(U) \to \mathcal{F}''(V)$. Thus this map is surjective and $\mathcal{F}''$ is flasque.

Now suppose that $\mathcal{F}$ is flasque. ∎

Proposition 4.2.4 Let X be a space and let $\mathcal{F}$ be a sheaf on X. If $\mathcal{F}$ is flasque then $\mathcal{F}$ is acyclic for $\Gamma(X, -)$. Explicitly, this means that

$$H^i(X, \mathcal{F}) = 0$$

for all $i > 0$.

Proof Since $\mathbf{Ab}(X)$ has enough injective objects, suppose that $\mathcal{I}$ is an injective object of $\mathbf{Ab}(X)$ such that $\mathcal{F}$ is a subobject of $\mathcal{I}$. Let $\mathcal{G}$ be the quotient sheaf $\mathcal{F}/\mathcal{I}$. Then there is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0$$

By lemma 3.2.2, $\mathcal{I}$ is flasque. By 3.2.3, $\mathcal{G}$ is flasque. By 3.2.3, there is an exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow 0$$

Since $\mathcal{I}$ is injective, we have that $H^i(X, \mathcal{I}) = 0$ for all $i > 0$. By passing to the long exact sequence in cohomology, we deduce that $H^1(X, \mathcal{F}) = 0$ and $H^i(X, \mathcal{F}) = H^{i-1}(X, \mathcal{G})$ for each $i \geq 2$. Applying the same method again gives $H^1(X, \mathcal{G}) = 0$. Thus by induction, we conclude that $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. ∎

Proposition 4.2.5 Let $(X, \mathcal{O}_X)$ be a ringed space. Then the derived functors of

$$\Gamma(X, -) : {}_{\mathcal{O}_X}\mathbf{Mod}(X) \to \mathbf{Ab}$$

coincide with the derived functors of $\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab}$.

Thus there is no distinction in considering the cohomology groups of sheaf of $\mathcal{O}_X$-modules and when considering them as simply a sheaf of abelian groups.

4.3 Čech Cohomology

Definition 4.3.1 (Čech Complex) Let X be a topological space and $\mathcal{U} = \{U_i | i \in I\}$ an open cover of X where I is an indexing set. For any $(i_0, \dots, i_k) \in I^{k+1}$, denote

$$U_{i_0, \dots, i_k} = U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_k}$$

Define for each k,

$$C^k(X, \mathcal{U}, \mathcal{F}) = \bigcap_{(i_0, \dots, i_k) \in I^{k+1}} \mathcal{F}(U_{i_0, \dots, i_k})$$

Furthermore, define a boundary map $d : C^k(X,\mathcal{U},\mathcal{F}) \to C^{k+1}(X,\mathcal{U},\mathcal{F})$ by

$$c_{i_0\dots,i_k} \overset{d}{\mapsto} \sum_{s=0}^{k+1}(-1)^s \text{res}(c_{i_0,\dots,\hat{i}_s,\dots,i_{k+1}})$$

Define the Čech complex to be $(C^\bullet(X,\mathcal{U},\mathcal{F}), d)$.

Lemma 4.3.2 For any space X and any open cover $\mathcal{U}$ of X, $(C^\bullet(X,\mathcal{U},\mathcal{F}), d)$ is indeed a chain complex.

Definition 4.3.3 (Čech Cohomology) Let $(C^\bullet(X,\mathcal{U},\mathcal{F}), d)$ be a Čech complex. Define the kth cohomology group of it to be

$$\check{\mathrm{H}}^k(X,\mathcal{U},\mathcal{F}) = \frac{\ker(C^k(X,\mathcal{U},\mathcal{F}) \to C^{k+1}(X,\mathcal{U},\mathcal{F}))}{\text{im}(C^{k-1}(X,\mathcal{U},\mathcal{F}) \to C^k(X,\mathcal{U},\mathcal{F}))} = H(C^\bullet(X,\mathcal{U},\mathcal{F}), d)$$

Lemma 4.3.4 For any Čech complex, we have that $\check{\mathrm{H}}^0(X,\mathcal{U},\mathcal{F}) = \mathcal{F}(X)$.

Proposition 4.3.5 Let X be a topological space and $\mathcal{U}$ an open cover of X. If the open sets $U_{i_0,\dots,i_k}$ satisfy that $H^k(U_{i_0,\dots,i_k},\mathcal{F}) = 0$ for all $k > 0$, then

$$H^k(X,\mathcal{F}) = \check{\mathrm{H}}^k(X,\mathcal{U},\mathcal{F})$$

5 Sites, Sieves and Grothendieck Topologies

5.1 Sites on a Category

We now want to generalize the notion of sheaves to general categories, not just spaces. However, there is no notion of open sets in an arbitrary category. Grothendieck realized that the essence of sheaves comes from the data of the open covers $U = \bigcup_{i\in I} U_i$ for each open set U in a space X.

Definition 5.1.1 (Covers) Let $\mathcal{C}$ be a category. Let $U \in \mathcal{C}$ be an object. A cover of U is a collection of morphisms
$$\{U_i \to U \mid i \in I\}$$

Definition 5.1.2 (Coverages (nLab)) Let $\mathcal{C}$ be a category. A coverage $\text{Cov}(\mathcal{C})$ of $\mathcal{C}$ consists a collection of covers for each $U \in \mathcal{C}$, such that the following are true.

- If $\{f_i : U_i \to U \mid i \in I\}$ is a cover of U and $g : V \to U$ is a morphism in $\mathcal{C}$, then there exists a cover $\{h_j : V_j \to V\}$ of V such that every $g \circ h_j : V_j \to U$ factors through the cover of U. Or in other words, for each $h_j : V_j \to V$, there exists U_i and $f_i : U_i \to U$ such that the following diagram commutes:
$$\begin{tikzcd} V_j \arrow[r, dashed, "k"] \arrow[d, "h_j"'] & U_i \arrow[d, "f_i"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

Definition 5.1.3 (Coverages (Stacks)) Let $\mathcal{C}$ be a category. A coverage $\text{Cov}(\mathcal{C})$ of $\mathcal{C}$ consists a collection of covers for each $U \in \mathcal{C}$, such that the following are true.

- If $V \to U$ is an isomorphism then $\{V \to U\} \in \text{Cov}(\mathcal{C})$
- If $\{U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$ and $\{V_{i,j} \to U_i \mid j \in J_i\} \in \text{Cov}(\mathcal{C})$ for each $i \in I$, then $\{V_{i,j} \to U_i \to U \mid i, j \in I\} \in \text{Cov}(\mathcal{C})$
- If $\{f_i : U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$ and $g : V \to U$ is a morphism, then the pullback
$$\begin{tikzcd} U_i \times_U V \arrow[r, dashed, "\text{proj.}"] \arrow[d, "\text{proj.}"'] & U_i \arrow[d, "g"] \\ V \arrow[r, "f_i"'] & U \end{tikzcd}$$
exists for all $i \in I$ and $\{U_i \times_U V \to U\} \in \text{Cov}(\mathcal{C})$.

Definition 5.1.4 (Sites (nLab)) A site is a category $\mathcal{C}$ together with a coverage $\text{Cov}(\mathcal{C})$.

Definition 5.1.5 (Continuous Functors) Let $\mathcal{C}$ and $\mathcal{D}$ be two sites. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. We say that F is continuous if the following are true.

- F preserves fiber products
- F sends covers to covers

6 Sheaf Cohomology of Sites

6.1 Sheaves on Sites

Definition 6.1.1 (Sheaves on Sites) Let $(\mathcal{C}, \text{Cov}(\mathcal{C}))$ be a site. Let $\mathcal{F}$ be a presheaf of sets on $\mathcal{C}$. We say that $\mathcal{F}$ if a sheaf on $\mathcal{C}$ if for all $\{f_i : U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$, the following diagram

$$\mathcal{F}(U) \xrightarrow{p} \prod_{i\in I} \mathcal{F}(U_i) \overset{q}{\underset{r}{\rightrightarrows}} \prod_{i,j\in I} \mathcal{F}(U_i \times_U U_j)$$

is an equalizer diagram, where p is the product of the maps $\mathcal{F}(f_i)$, q is the product of the maps $\mathcal{F}(\text{proj}_i : U_i \times_U U_j \to U_i)$ and r is the product of the maps $\mathcal{F}(\text{proj}_j : U_i \times_U U_j \to U_j)$.

Definition 6.1.2 (Abelian Sheaves) Let $\mathcal{C}$ be a site. Define the category of abelian sheaves

$$\mathbf{Ab}(\mathcal{C})$$

over $\mathcal{C}$ to consist of the following data.

- The objects are the abelian sheaves on $\mathcal{C}$
- The morphisms are the natural transformations of functors
- Composition is given by the composition of natural transformations

Theorem 6.1.3 Let $\mathcal{C}$ be a site. Then $\mathbf{Ab}(\mathcal{C})$ has enough injectives.

TBA: category of abelian sheaves on $\mathcal{C}$ has enough injectives

TBA: A bunch of adjunctions analogous to that of sheaves.

6.2 Sheaf Cohomology of Sites

Definition 6.2.1 (Global Sections Functor) Let $\mathcal{C}$ be a site. Define the global section functor

$$\Gamma(-, \mathcal{C}) : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$$

by the following data.

- A sheaf $\mathcal{F}$ on $\mathcal{C}$ is sent to $\Gamma(\mathcal{F}, \mathcal{C}) = \mathcal{F}(\mathcal{C})$.
- For a morphism of sheaves $\phi : \mathcal{F} \to \mathcal{G}$, the global section functor sends it to the group homomorphism $\mathcal{F}(\mathcal{C}) \to \mathcal{G}(\mathcal{C})$

Proposition 6.2.2 Let $\mathcal{C}$ be a site. Then the global section functor $\Gamma(-, \mathcal{C}) : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$ is left exact.

Definition 6.2.3 (Sheaf Cohomology of Sites) Let $\mathcal{C}$ be a site. Define the sheaf cohomology functor of $\mathcal{C}$ to be

$$H^k(\mathcal{C}, -) = R^k\Gamma : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$$

the right derived functors of $\Gamma(-, \mathcal{C})$.

7 Etale Cohomology

To be removed:

Definition 7.0.1 ((Small) Etale Sites) Let X be a scheme. The etale site X_{et} over X consists of the following data.

- The underlying category is the full subcategory $\mathbf{Sch}/X$ of schemes over X consisting of etale morphisms $U \to X$
- The coverage consists of the surjective families of etale morphisms $\{U_i \to U \mid i \in I\}$.

Finite etale site, the flat site, the big etale site

Definition 7.0.2 (Etale Cohomology) Let X be a scheme. Define the etale cohomology of X to be the sheaf cohomology of the category X_{et}. Explicitly, this means that

$$H^k_{\text{et}}(X, -) = H^k(X_{\text{et}}, -) : \mathbf{Ab}(X_{\text{et}}) \to \mathbf{Ab}$$